

Analog and Digital Data

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 - take on continuous values.
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Analog and Digital Data

☐ **Example:**

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- On the other hand, a digital clock that reports the hours and the minutes will change suddenly from 8:05 to 8:06.





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Analog and Digital Signals

- Like the data they represent, signals can be analog or digital.
- An “**Analog signal**” has infinitely many levels of intensity in a period of time.
- “**Digital signals**” can have only a limited number of defined values.

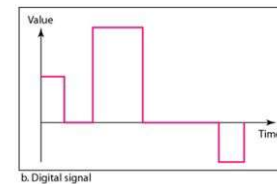
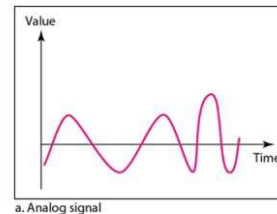
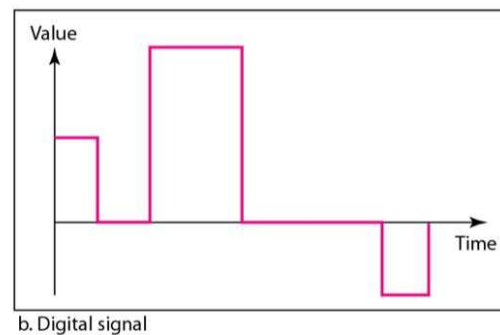
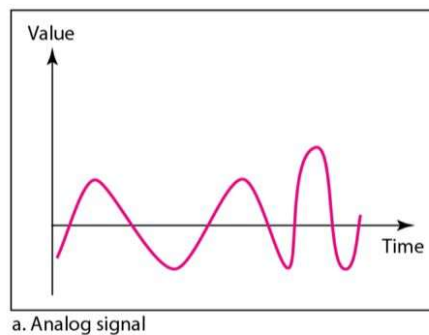


Figure 3.1 *Comparison of analog and digital signals*



Q.2

Unguided media -Introduction

- ▶ The section of the electromagnetic spectrum defined as radio waves and microwaves is divided into **eight ranges**, called **bands**.

Table 7.4 Bands

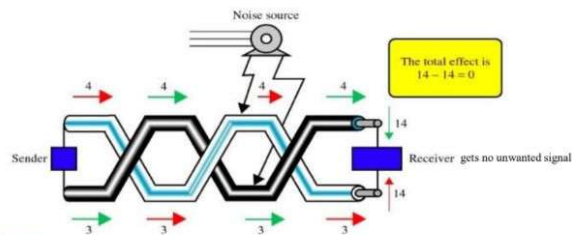
<i>Band</i>	<i>Range</i>	<i>Propagation</i>	<i>Application</i>
very low frequency (VLF)	3–30 kHz	Ground	Long-range radio navigation
low frequency (LF)	30–300 kHz	Ground	Radio beacons and navigational locators
middle frequency (MF)	300 kHz–3 MHz	Sky	AM radio
high frequency (HF)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft
very high frequency (VHF)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
ultrahigh frequency (UHF)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
superhigh frequency (SHF)	3–30 GHz	Line-of-sight	Satellite
extremely high frequency (EHF)	30–300 GHz	Line-of-sight	Radar, satellite

Q.3

1. TWISTED CABLE:

Twisted Pair Cable

- ▶ Consists of 2 copper conductors (normally copper), each having its own plastic insulation.
- ▶ Two wires are twisted around each other.
- ▶ One wire carries electrical signals and the other is ground reference.
- ▶ The receiver uses the difference between the two.
- ▶ Typically twisted pair is installed in building telephone wiring.
- ▶ Twisting is done to reduce the effect of noise on the pair. Each wire is closer to the noise source for half time and farther for the other half.



Transmission Media

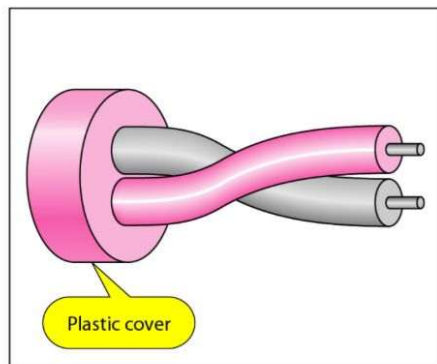
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Twisted Pair Cable

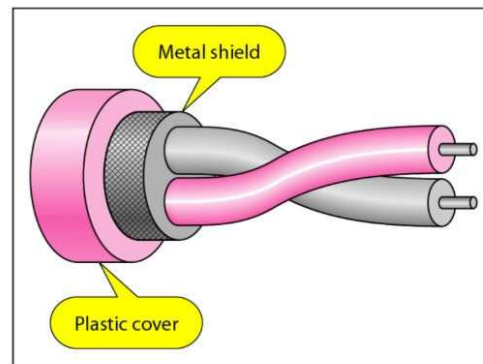
▶ **Types:**

1. Unshielded twisted pair
2. Shielded twisted pair

UTP(Unshielded twisted pair) and STP(Shielded twisted pair) cables



a. UTP



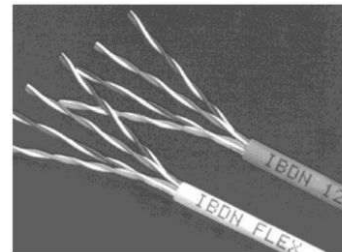
b. STP

7.14

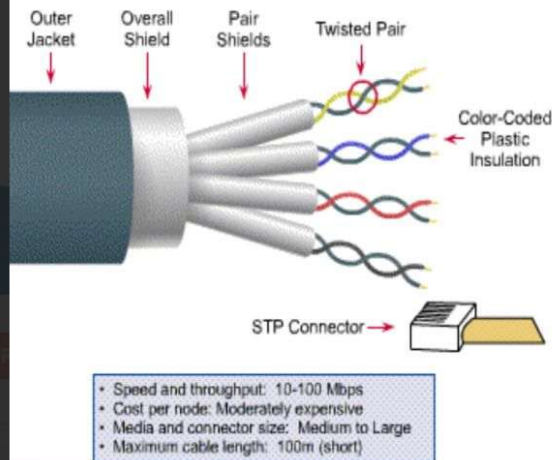
Unshielded Twisted Pair (UTP)

UTP

- Consists of 4 pairs (8 wires) of insulated copper wires typically about 1 mm thick.
- The wires are twisted together in a helical form.
- Flexible and cheap cable.
- Categories: CAT 3, CAT 4, CAT 5, Enhanced CAT 5, CAT 6 and now CAT 7.
- UTP comes in several categories that are based on the cable quality.



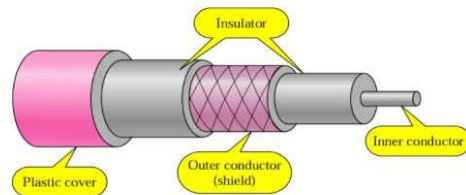
Shielded Twisted Pair (STP)



- STP cable has a **metal foil** or **braided-mesh** covering that enhances each pair of insulated conductors.
- The metal casing prevents the penetration of electromagnetic noise.
- Materials and manufacturing requirements make STP more **expensive** than UTP but **less susceptible to noise**

Coaxial cable

- Coaxial cable is a copper-cored cable surrounded by a heavy shielding and is used to connect computers in a network.
- The outer metallic wrapping serves both as a shield against noise and as the second conductor which completes the circuit.
- High bandwidth but lossy channel.
- Repeater is used to regenerate the weakened signals.



Coaxial cable

► Standards

Table 7.2 *Categories of coaxial cables*

Category	Impedance	Use
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

RG - Radio Government rating

Coaxial cable

Coaxial Cable Connectors

- ▶ To connect coaxial cable to devices, it is necessary to use coaxial connectors.
- ▶ The most common type of connector is the Bayone-Neill-Concelman, or **BNC connectors**.

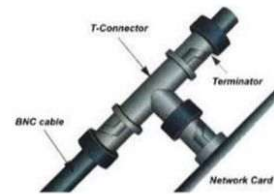
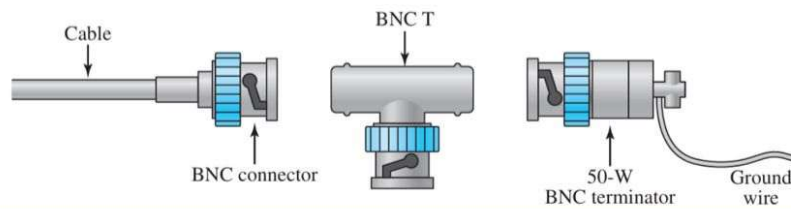


Figure 7.8 BNC connectors



Fiber Optics

- ▶ Made of glass or plastic, transmits signals in the form of light
- ▶ Light pulse can be used to signal a 1-bit and the absence of a pulse a 0 bit.

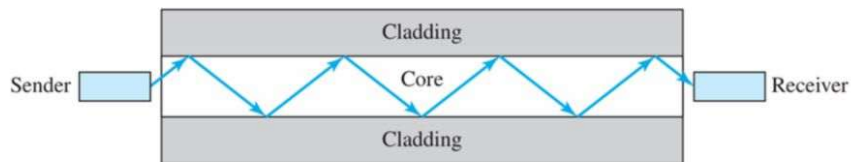
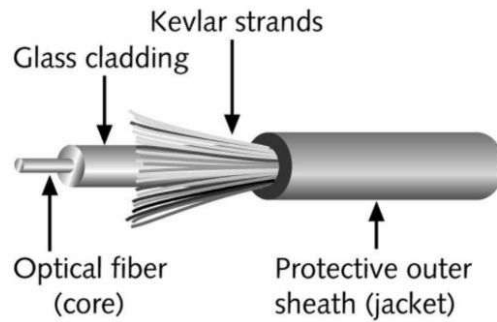
Fiber Optics

- ▶ Optical transmission has 3 components:
 1. Transmission media (ultra thin glass fiber)
 2. Light source (LED/laser diode) both emit light pulses when an electric current is applied.
 3. Detector(Photodiode) : Generates an electric pulse when light falls on it.
- ▶ By attaching an LED/laser to one end of the optical fiber and a photodiode to the other, we can create a unidirectional data transmission system that →
 1. Accepts an electrical signal, converts it to light pulses and transmits these light pulses.
 2. Reconverts the output to an electrical signal at the receiving end.

Fiber Optics

Structure

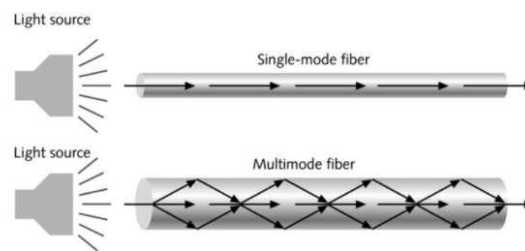
- Contains one or several glass fibers at its **core**
- Surrounding the fibers is a layer called **cladding**



Fiber Optics

Types of fiber Optics

- Single-mode fiber**
 - Carries light pulses along single path
 - Uses Laser Light Source
- Multimode fiber**
 - Many pulses of light generated by LED travel at different angles



Fiber Optics

Connectors:

1. **Subscriber channel (SC) connector**

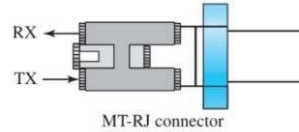
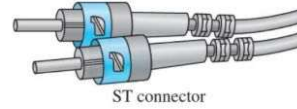
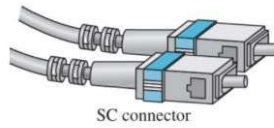
■ is used for cable TV

2. **Straight-tip (ST) connector**

■ is used for connecting cable to networking devices

1. **MT-RJ**

■ is a connector that is the same size as RJ45



4.RADIO WAVES:

Radio Waves (3KHz – 1 GHz)

- ▶ Radio waves are **omnidirectional**. When an antenna transmits radio waves, they are propagated in all directions. This means that the sending and receiving antennas do not have to be aligned. A sending antenna send waves that can be received by any receiving antenna.

Omnidirectional Antenna for Radio Waves

- ▶ Radio waves use omnidirectional antennas that send out signals in all directions. They follow the curvature of the earth.
- ▶ The radio wave **band is relatively narrow**, just under 1 GHz. Hence, sub-bands are narrower → low data rate for digital communications!



Radio Waves

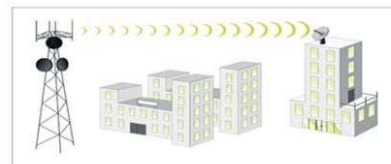
Applications of Radio Waves

- ▶ The omnidirectional characteristics of radio waves make them useful for multicasting in which there is one sender but many receivers.
- ▶ AM(Amplitude modulation) and FM(Frequency modulation) radio, television, maritime radio, cordless phones, and paging are examples of multicasting.

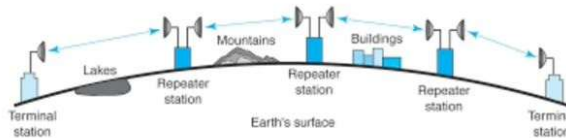
5. MICROWAVES

Microwaves (1 – 300 GHz)

- ▶ Microwave are unidirectional.
- ▶ Transmissions can be can be narrowly focused.
- ▶ Require **line of sight transmission** and reception equipment.
- ▶ Transmission/reception equipment if at a height of 100 m, distance between them can be max 82km.



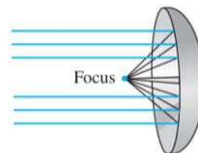
- ▶ To increase the distance → system of repeaters can be used



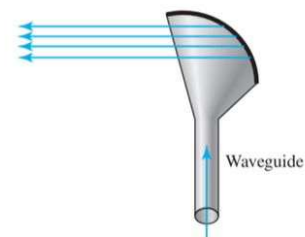
Microwaves

- ▶ Very high-frequency microwaves cannot penetrate walls which is a disadvantage if receivers are inside buildings.
- ▶ The microwave **band is relatively wide**, almost 299 GHz. Therefore, wider sub-bands can be assigned, and a high data rate is possible.

Unidirectional antennas :



a. Parabolic dish antenna



b. Horn antenna

Applications :

Cellular telephones,
satellite communication &
Wireless LANs

6. INFRARED WAVES:

Infrared

- ▶ Infrared waves, with frequencies from 300 GHz to 400 THz, can be used for **short-range communication** using line-of-sight propagation.
- ▶ Infrared waves, having high frequencies, **cannot penetrate walls**.
- ▶ This **advantageous characteristic** prevents interference between one system and another, a short-range communication system in one room cannot be affected by another system in the next room.
- ▶ When we use infrared remote control, we do not interfere with the use of the remote by our neighbours.
- ▶ However, this same characteristic makes infrared signals useless for long-range communication!
- ▶ In addition, we **cannot use** infrared waves **outside a building** because the sun's rays contain infrared waves that can interfere with the communication!

Infrared

Applications of Infrared Waves

- ▶ The infrared band, almost 400 THz, has an excellent potential for data transmission.
- ▶ Such a wide bandwidth can be used to transmit digital data with a very high data rate.

